

Precise Construction of Helical Unimolecular Micelles from Poly(phenylacetylene)s for Tunable Circularly Polarized Luminescence and Sensing

Zhenqian Zhang,^{1,‡} Yuhang Wei,^{1,‡} Xiaoguang Qiao,^{1,3,‡} Junle Zhang,⁴ Mengjie Zhou,¹ Yanjie He,¹ Wenjie Zhang,¹ Zhe Cui,¹ Ge Shi,^{1,*} and Xinchang Pang^{1,2*}

¹Henan Joint International Research Laboratory of Living Polymerizations and Functional Nanomaterials, Henan Key Laboratory of Advanced Nylon Materials and Application, School of Materials Science and Engineering, Zhengzhou University, Zhengzhou 450001, China

²School of Materials Science and Engineering, Henan University of Science and Technology, Luoyang 471023, China

³College of Materials Engineering; Henan International Joint Laboratory of Rare Earth Composite Materials, Henan University of Engineering, Zhengzhou 451191, China

⁴Faculty of Engineering, Huanghe Science and Technology College, Zhengzhou 450063, China

Corresponding Author

E-mail: geshizzu@zzu.edu.cn

Pangxinchang1980@163.com

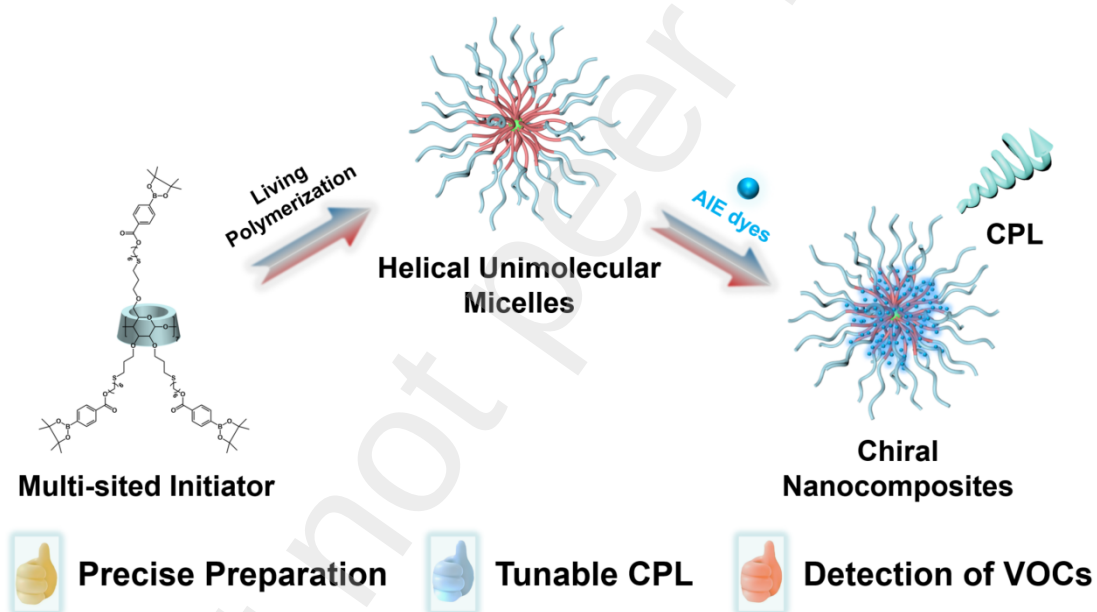
[*] To whom correspondence should be addressed.

[‡] These authors contributed equally to this work.

For Table of Contents Used Only

Precise Construction of Helical Unimolecular Micelles from Poly(phenylacetylene)s for Tunable Circularly Polarized Luminescence and Sensing

Zhenqian Zhang,^{1,‡} Yuhang Wei,^{1,‡} Xiaoguang Qiao,^{1,3,‡} Junle Zhang,⁴ Mengjie Zhou,¹ Yanjie He,¹ Wenjie Zhang,¹ Zhe Cui,¹ Ge Shi,^{1,*} and Xinchang Pang^{1,2*}



Abstract

Chiral nanocomposites have attracted wide attention due to their fascinating size-dependent characteristics. Such nanomaterials could be precisely synthesized by templates from random-coiled unimolecular micelles with poor chiroptical activities. Helical polymers possessing secondary chirality usually displayed the stronger chiral induction than random-coiled polymers. However, the controllable preparation of helical unimolecular micelles still remains challenging, which hindered the investigation into the effect of morphology and size on chiroptical properties. In this study, multi-armed star-shaped poly(phenylacetylene)s (PPAs) with target composition, molecular weight and size, were first precisely prepared from the novel multi-sited initiator via living polymerization. Based on acid-base interactions, chiral transfer from PPAs to various dyes was accomplished in composite membranes and suspension. Notably, the chiral nanocomposites consisting of star-shaped PPAs displayed enhanced helical stability, Cotton effect and circularly polarized luminescence (CPL) compared with linear analogues. Furthermore, this system displayed the triple-mode (photoluminescence, circular dichroism and CPL) detection towards the volatile organic compound for environmental monitoring.

Keywords: Poly(phenylacetylene)s; Unimolecular micelles; Chiral induction; Circularly polarized luminescence; Precise synthesis

Introduction

Chirality, as one of the most ubiquitous structural properties in nature, has been found the widespread applications including biomedicine^{1, 2}, asymmetric catalysis³⁻⁵, and optoelectronic devices⁶⁻⁹. For the preparation of chiral nanomaterials, the template method as highly selective methodology was widely utilized¹⁰. Such methods were classified into hard-template and soft-template approaches. The hard templates derived from metal-organic frameworks¹¹ and silica^{12, 13}, could exhibit enhanced structural stability. Meanwhile, soft templates (such as DNA¹⁴, proteins¹⁵, and polymers¹⁶) display the satisfactory chiral tunability and stimuli-responsiveness to the external environmental factors.

Compared to other soft templates, polymer-based templates usually exhibit satisfactory designability, and their corresponding self-assembly behavior paved the way for chiral nanomaterials¹⁷⁻¹⁹. Such assemblies of linear polymers required specific conditions and exhibits poor structural stability and weak size-controllability. Notably, star-shaped unimolecular micelles with each arm covalently linked to the core-structure, showed structural stability even under harsh conditions. By living polymerization, such star-shaped unimolecular micelles act as the dimension- and size-controllable nanoreactors, representing an ideal platform for the precise construction of nanomaterials^{20, 21}. If chiral monomers were employed, the corresponding chiral templates achieved the production of chiral nanocomposites with target size and morphology^{22, 23}. However, the previous work mainly focused on polymers with point chirality and random-coiled mainchain. Thus, the chiral transfer from templates to nanostructures was not efficient enough.

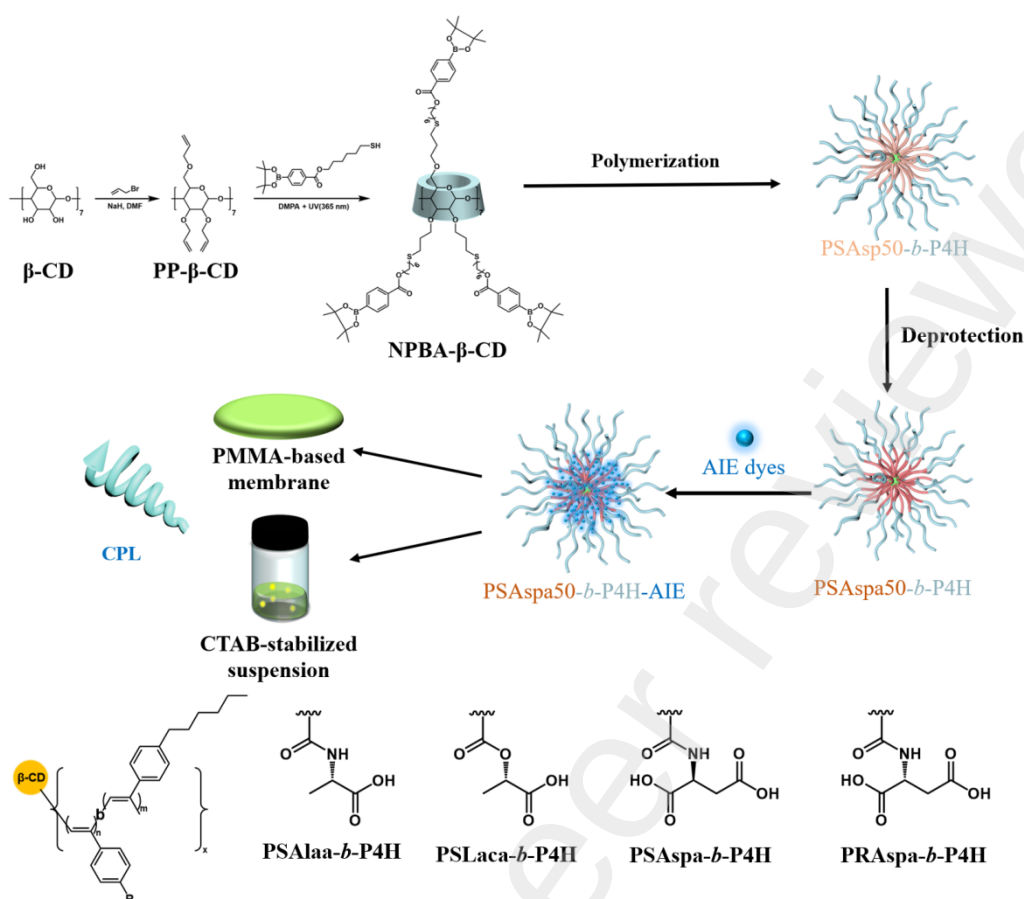
Helical polymers containing secondary chirality, are believed to outperform random-coiled polymers in the field of chiral induction, enantio-separation, and asymmetric catalysis²⁴⁻²⁶. The poly(phenylacetylene)s (PPAs) is one of the representative helical polymers, which have attracted significant attention due to their dynamic helicity, living polymerization, and application potential²⁷⁻²⁹. However, only chiral three-armed PPAs could be obtained via living polymerization, displaying

distinctive Cotton effect and less intermolecular entanglement compared with linear PPAs^{30, 31}. And the achiral multi-armed (above three) PPAs were prepared with poor controllability over polymerization and structures³². As a result, the precise construction of multi-armed star-shaped PPA-based templates is still challenging and the effect on chiral transfer from PPA-based templates to other components remains ambiguous.

In this study, chiral multi-armed star-shaped block PPAs were precisely prepared, through living polymerization from multi-sited initiators. To the best of our knowledge, this is the first reported chiral multi-armed PPA-based unimolecular micelles possessing target molecular weight and size. The relationship between chemical structures, morphology, molecular weight and chiroptical activities of PPAs was figured out, while the chiral induction from PPA-based templates towards aggregation-induced emission (AIE) dyes and circularly polarized luminescence (CPL) were also systematically studied. Furthermore, the CPL-system exhibited higher sensitivity towards volatile organic compounds than the fluorescent sensor, indicating the application potential of environmental monitoring.

Results and Discussion

The initiator with multiple sites plays an important role in the synthesis of PPA-based unimolecular micelles. Firstly, the thiol derivative and allyl-decorated β -cyclodextrin (PP- β -CD) was synthesized by condensation and nucleophilic substitution, respectively (Figure S1-S2). Based on click reaction, the phenylboronic acid pinacol ester was successfully introduced into β -cyclodextrin as the multi-sited initiator NPBA- β -CD. The grafting yield of NPBA- β -CD was estimated as 90% (Figure S3). Besides, various chiral (MSAla, MSLac, MSAsp and MRAsp) and achiral phenylacetylene derivatives (MT) bearing *tert*-butyl esters were also prepared with high yields (Figure S4-S7).



Scheme 1. Synthetic routes of initiator, star-shaped polymers, and chiral composites as PMMA-based membrane and CTAB-stabilized suspension.

To prove living characteristics of polymerization, we selected NPBA- β -CD as the initiator and phenylacetylene as the monomer. When the feed ratio ($[M] : [I]$) was 50 : 1, star-shaped PPAs were obtained at different polymerization time. The number-averaged molecular weight (M_n) was increased from 18 to 71 kDa with extended time (Table S1). The polydispersity index (PDI) of these PPAs were below 1.15, indicating the satisfactory controllability. GPC curves exhibited narrow monomodal distribution, confirming the absence of cross-linked structures or oligomers (Figure S8). Moreover, the reaction was conformed to the first-order kinetic polymerization (Figure S9), which is conducive to the precise preparation of PPA-based unimolecular micelles. Thus, a series of star-shaped homopolymers bearing different pendants were obtained with high yields (Table 1), named as PPA50, PSAla50, P4H50, PT50 and PT100 (the number denoted the target degree of polymerization from each polymeric arm). Increasing the

[M] : [I] from 50 to 100, the PT100 with larger M_n was observed at less retention time, but the PDI was still maintained as 1.12 (Figure S10).

Achiral 4-hexyl-1-ethynylbenzene (M4H) was utilized as the second monomer (Scheme 1), to improve the solubility of PPAs. Compared with homopolymers, the M_n of corresponding block copolymers were further increased (Table 1, Entry 6-10). According to the $^1\text{H-NMR}$ spectra, the *cis*-structure contents of these star-shaped PPAs were all above 90% (Figure S11-S14), which accorded with the stereo-control of Rh-based catalysts³³. To confirm that monomers were completely consumed in the first step of block copolymer synthesis, the model polymerization from MSAsp was conducted (Figure S15). And the vanished signal of ethynyl proton proven that the conversion of monomers was nearly 100%, before the addition of second monomer²⁴.

Table 1. Polymerization results of star-shaped PPAs^a

Entry	Name	[M]: [I]	Monomer	Conv. (%) ^c	$M_{n,\text{GPC}}(\text{kDa})^d$	M_w/M_n^d
1	PPA50	50	MPA	96	71	1.15
2	PSAla50	50	MSAla	89	95	1.06
3	P4H50	50	M4H	94	143	1.13
4	PT50	50	MT	95	116	1.12
5	PT100	100	MT	95	239	1.12
6	PSLac50- <i>b</i> -P4H ^b	50	MSLac	87	271	1.13
7	PSAla50- <i>b</i> -P4H ^b	50	MSAla	82	226	1.13
8	PSAsp50- <i>b</i> -P4H ^b	50	MSAsp	71	208	1.08
9	PRAsp50- <i>b</i> -P4H ^b	50	MRAsp	71	205	1.07
10	PSAsp100- <i>b</i> -P4H ^b	100	MSAsp	65	356	1.34

^a Reaction conditions: the polymerization was conducted using NPBA- β -CD as the initiator rhodium-diphenylacetylene-triphenylphosphine complex as the catalyst under basic conditions in THF at 30°C for 1 h. ^b The second block was introduced by sequential addition of M4H monomer, followed by continued polymerization at 30 °C for 3 h. ^c The monomer conversion was determined by gravimetric analysis of the polymer. ^d The molecular weight (M_n) and polydispersity index ($\text{PDI} = M_w/M_n$) of the polymers were determined by gel permeation chromatography (GPC) using THF as eluent, with polystyrene standards for calibration.

Chiral induction based on acid-base interaction between carboxyl derivatives and amines, was proven as an effective mode³⁴. And carboxyl-modified PPAs were

obtained by the treatment of trifluoroacetic acid. After deprotection (Figure S16a), the appearance of characteristic carboxyl proton peaks at 12.87 ppm, and the absence of *tert*-butyl ester (1.22-1.79 ppm) were observed. Besides, the carbonyl stretching vibration at 1732 cm^{-1} was shifted to 1647 cm^{-1} , accompanied by the emergence of the broad O-H stretching band around 3000-3500 cm^{-1} (Figure S16b). The deprotection process did not change other functional groups (Figure S17-S19), suggesting the high selectivity of deprotection. From TEM characterization by negative staining of uranyl acetate (Figure S20), uniform star-shaped PPAs displayed the averaged size as 80 nm (PSAspa50-*b*-P4H) and 186 nm (PSAspa100-*b*-P4H). Such tendency was also observed from the DLS spectra (85 nm and 200 nm). In addition, the size distribution of PSAspa100-*b*-P4H was wider than PSAspa50-*b*-P4H, which was probably owing to the larger PDI.

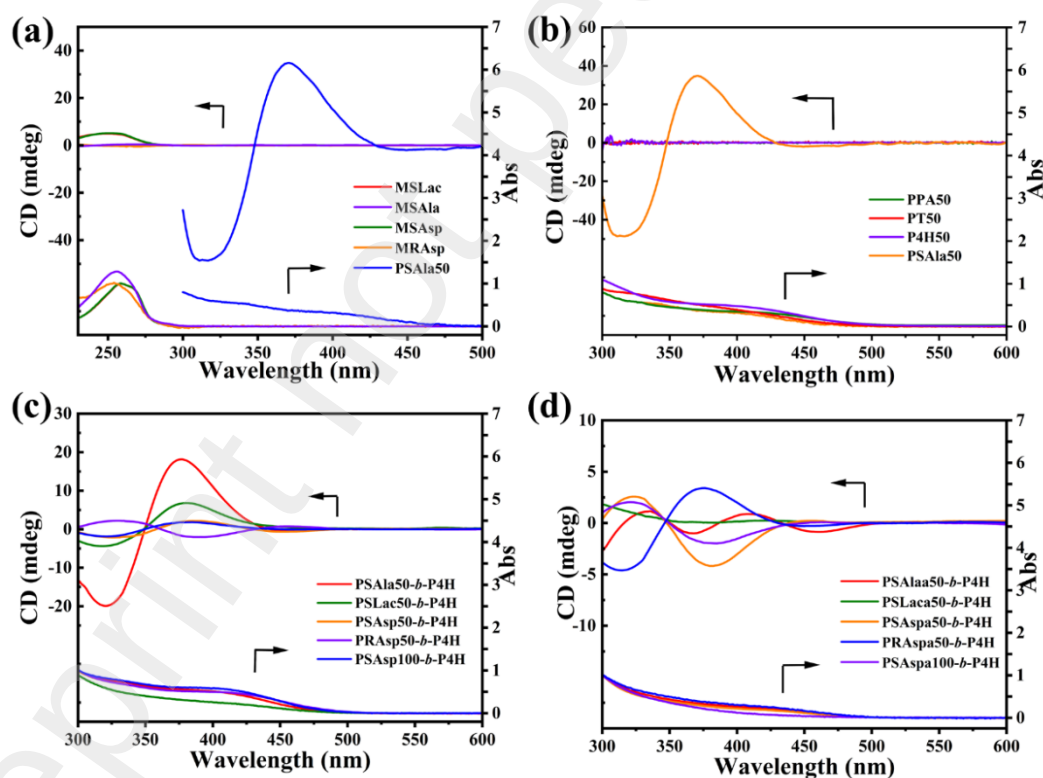


Figure 1. (a) UV-Vis absorption and CD spectra of monomers and PSAla50, (b) star-shaped homopolymers, (c) star-shaped block copolymers, and (d) polymers after deprotection in THF.

To investigate the structural effect on circular dichroism (CD), the CD measurement of monomers and polymers was conducted. For the chiral monomer MSAla in Figure 1a, only slight CD signal was observed below 300 nm. After polymerization, the corresponding star-shaped PSAla50 displayed new absorption above 300 nm, attributing to the conjugated backbone. Meanwhile, the enhanced Cotton effect in the range of 300-400 nm implied the formation of predominantly one-handed helical main-chain ($g_{CD} = 2.9 \times 10^{-3}$). For the achiral homopolymers, the silent-CD suggested that the helical conformation was induced by chiral pendants (Figure 1b).

After the introduction of achiral P4H block (Figure 1c), copolymer PSAla50-*b*-P4H showed weaker Cotton effect ($g_{CD} = 9.9 \times 10^{-4}$) than PSAla50, due to the decreased content of chiral repeating units. When chiral units were transformed to lactic acid or aspartic acid, the weaker chiroptical activities of PSLac50-*b*-P4H ($g_{CD} = 6.9 \times 10^{-4}$) and PSAsp50-*b*-P4H ($g_{CD} = 1.3 \times 10^{-4}$) were observed. The enhanced CD of PSAla50-*b*-P4H may be attributed to: (1) the better chiral transfer from pendant to mainchain as the stronger optical activity of MSAla; (2) the helical conformation was stabilized by intramolecular hydrogen bonds in PSAla50-*b*-P4H^{35, 36}. Increasing the degree of polymerization, PSAsp100-*b*-P4H exhibited similar CD spectrum to PSAsp50-*b*-P4H, indicating the negligible effect of molecular weight on chiroptical properties. When the configuration of pendants was changed (PSAsp50-*b*-P4H and PRAsp50-*b*-P4H), the mirror-imaged Cotton spectra suggested that the optical activity of backbones originates from the configuration of side groups³¹. After deprotection, the CD intensity of star-shaped copolymers was further decreased (PSAspa50-*b*-P4H, $g_{CD} = -4.8 \times 10^{-4}$; PSAlaa50-*b*-P4H, $g_{CD} = -1.1 \times 10^{-4}$; PSLaca50-*b*-P4H, $g_{CD} = -5.3 \times 10^{-5}$). Without the group of *tert*-butyl esters, the less steric hinderance of pendants was believed to destroy the helical screw-sense in mainchain (Figure 1d)³⁷.

Interestingly, the morphology of PPAs displayed the obvious influence on chiroptical activities. Linear PPAs were also prepared by living polymerization from the mono-sited initiator (Figure S22 & Table S2). In Figure S23a, the dissymmetry factor of star-shaped PSAla50 ($g_{CD} = 2.9 \times 10^{-3}$) obviously exceeded its linear analogue PSAla50-L ($g_{CD} = 1.2 \times 10^{-3}$). That trend also accorded with block copolymers (Figure

S23b). For the star-shaped PPAs, one chain-end of each polymer arm was covalently attached to the β -CD core. And the reduced flexibility of PPA chain could lead to the stronger chiral amplification in star-shaped polymers²⁹.

As shown in Figure 2, the polymers exhibited dynamic helical characteristics under variable temperatures. The quasi-first-order reaction rate constants of PSAla50 and linear PSAla50-L at variable temperatures were calculated. And the activation energy of helix inversion (E_a) was estimated according to the following equation, where R is the gas constant ($8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$)³⁸.

$$\ln k_{rac} = -(E_a/R) \times T^{-1} + b$$

The Arrhenius kinetic (Figure 3c) plot revealed that the E_a of star-shaped PSAla50 (82.2 kJ/mol) was higher than that of linear PSAla50-L (75.8 kJ/mol). As a result, the higher distribution density and covalent attachment of polymeric arms in star-shaped PPAs, could strengthen the stability of helical conformation to some extent.

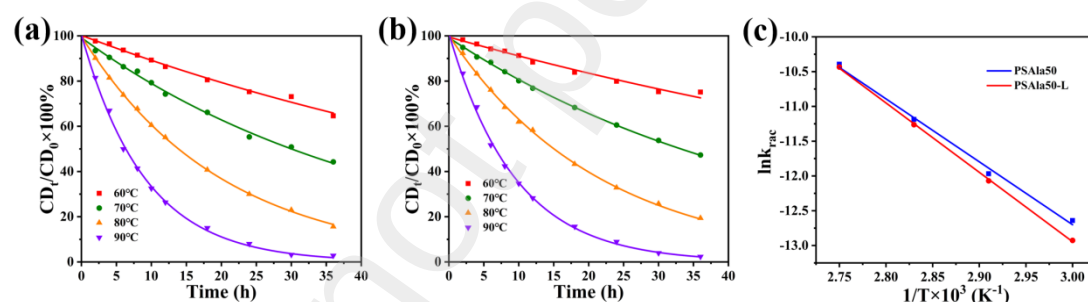


Figure 2. Time-dependent CD intensity at 394 nm of (a) PSAla50-L, (b) PSAla50 in DMA at various temperatures (60–90 °C). (c) The temperature dependence of the first order rate constants obtained at 60–90 °C for PSAla50-L and PSAla50 in DMA.

The enhanced CD and stability of star-shaped PPAs were propitious to the chiral induction from PPA-based templates to guest molecules. Four AIE dyes were chosen, named as 1-AE, 4-AE, 1-CE, and 4-CE (Scheme S4). When 1-AE was complexed with PSAspa50-*b*-P4H (Figure S24), the NMR signal of amino groups (4.62–5.16 ppm) became broader, suggesting the formation of acid-base interaction. And the AIE property of these chiral composites could be obtained as PMMA-based membrane and

CTAB-stabilized suspension (Figure S25).

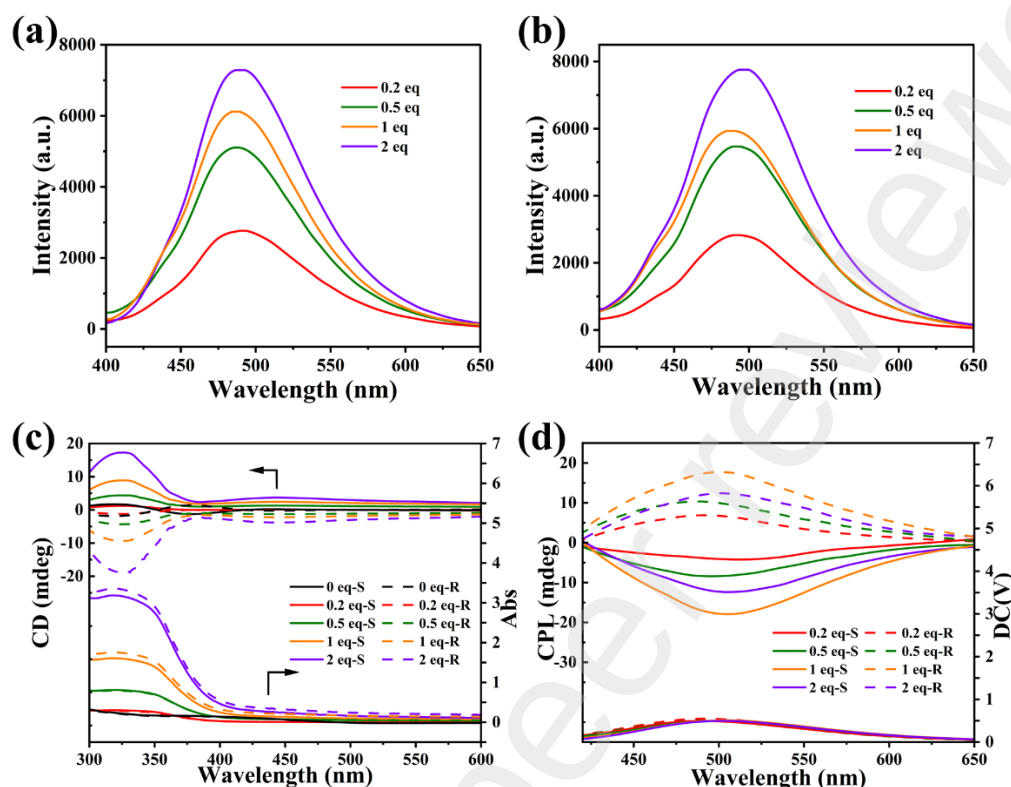


Figure 3. Fluorescence (a, b), UV-Vis absorption and CD (c), along with (d) CPL spectra of PSAspa-P4H-1AE and PRAspa-P4H-1AE as PMMA-based membranes.

For the membrane, the amount effect of dyes was studied. When the mole ration between 1-AE and carboxyl groups of PSAspa50-*b*-P4H was increased from 0 to 2 eq, the photoluminescence (PL) and CD around 325 nm belonging to the AIE dye were both enhanced (Figure 3a-b). This phenomenon may be attributed to the more 1-AE molecules interact with templates, resulting in the stronger chiral induction. The CPL property also exhibited concentration-dependent behavior (Figure 3d). The CPL intensity and g_{lum} increased in the range of 0.2-1 eq (0.2 eq as -5.9×10^{-4} ; 0.5 eq as -1.2×10^{-3} ; 1 eq as -2.5×10^{-3}), but decreased at higher loadings (2 eq as -1.7×10^{-3}). This result could be rationalized by the origin of CPL activities, as chiral induction towards 1-AE³⁹⁻⁴¹ and matching rules⁴². In the range of 0.2-1 eq, the acid-base equilibrium allowed more 1-AE to interact with helical templates, where the chiral induction and

CPL were strengthened simultaneously. Under higher ratios (1-2 eq), the excess 1-AE tended to form achiral aggregates, thus displaying photoluminescence (PL) not CPL and decreasing g_{lum} was measured. Actually, the g_{lum} values of this system are comparable to chiral molecules and helical polymers^{43, 44}. And the mirror-imaged CD and CPL between PSAspa-P4H-1AE and PRAspa-P4H-1AE, further indicated the key role of PPA-based templates during chiral induction. To rule out the contribution from linear polarization effects, we conducted angle-resolved measurements on the composite film containing PSAspa-P4H-1AE. As illustrated in Figure S26-S27, both the CD and CPL spectra exhibited negligible variations across all tested orientations. These results indicated that the chiroptical activities were slightly affected by the linear dichroism and linear birefringence phenomenon.

For the structural effect on the optical activity of chiral membranes, different PPA-based unimolecular micelles were utilized. As the content of dyes increased (0-2 eq.), the membranes prepared from different templates still showed the similar concentration-dependent behavior (Figure S28). And the optimal loading of 1-AE was equivalent to the carboxyl groups of templates. After the addition of 1 eq. 1-AE (Figure S29a), it was found that PPAs derived from aspartic acids possessed the strongest CD (PSAspa50-*b*-P4H, $g_{CD} = 1.7 \times 10^{-4}$; PSLaca50-*b*-P4H, $g_{CD} = 1.3 \times 10^{-4}$; PSAlaa50-*b*-P4H, $g_{CD} = 1.0 \times 10^{-4}$). This phenomenon may be attributed to the distinctive steric effect and chiral induction ability of such pendant. In Figure S29d, the CPL of PSAspa50-P4H composites ($g_{lum} = -2.5 \times 10^{-3}$) outperformed others (PSLaca50-*b*-P4H, $g_{lum} = -1.2 \times 10^{-3}$; PSAlaa50-*b*-P4H, $g_{lum} = -1.5 \times 10^{-3}$), further proving the strengthening chiral transfer from PSAspa50-*b*-P4H to 1-AE. If larger template PSAspa100-*b*-P4H was utilized, the CD and CPL intensity displayed tiny change compared with PSAspa50-*b*-P4H (Figure S29c&f). Meanwhile, the chiral membrane consisting of 1-AE and linear PSAspa50-*b*-P4H-L was prepared. Interestingly, the star-shaped composite ($g_{CD} = 1.7 \times 10^{-4}$, $g_{lum} = -2.5 \times 10^{-3}$) still outperformed the linear analogue ($g_{CD} = 1.5 \times 10^{-4}$, $g_{lum} = -1.8 \times 10^{-3}$). Generally, multi-armed star-shaped PPAs was believed to be the better chiral soft-templates than linear PPAs, as enhanced chiroptical activities and chiral induction. Besides, this chiral transfer and concentration dependence (Figure S30)

could also be achieved for other carboxyl-decorated dyes as 1-CE ($g_{CD} = 1.3 \times 10^{-4}$, $g_{lum} = -1.2 \times 10^{-3}$) and 4-CE ($g_{CD} = 8.9 \times 10^{-5}$, $g_{lum} = -1.1 \times 10^{-3}$). But for 4-CE possessing four amine groups, the resulting complex with PSAspa50-b-P4H exhibited poor dispersity as precipitation in the THF solution, which hindered the preparation of homogeneous PMMA membranes.

Apart from the chiral membrane, the chiral suspension was also investigated. In detailed, the THF solution of PPA-based unimolecular micelles and 1-AE was injected into the aqueous solution of cetyltrimethylammonium bromide (CTAB) surfactant. Following by heating at 353 K to evaporate THF solvent, the CPL suspension was obtained with the highest g_{lum} as -2.5×10^{-3} (Figure S31-S33). For 1-CE and 4-CE, the relative strong solvation of AIE dyes went against the photoluminescence and the CPL was hardly characterized (Figure S34).

In addition, volatile organic compounds (VOCs) with low boiling points and vapor pressure are the representative environmental pollutants, which would cause headaches, nausea, vomiting, and other disease in our living room^{45, 46}. Organic n-propylamine was selected as the analyte for VOC detection. Under the fumigation of n-propylamines, the decreased PL signals of PMMA-based membranes was observed with time (Figure 4a). That was because n-propylamines acted as the plasticizer to enhance the molecular motion of dyes and weaken their AIE properties in composite membranes. Besides, n-propylamines could compete with AIE dyes on the acid-base interactions between amines and PPAs. And the induced Cotton effect of dyes around 325 nm was weakened, while the characteristic absorption peak still maintained (Figure 4b). Actually, the CPL performance was influenced by both AIE and acid-base interactions (Figure 4c). Thus, CPL-system exhibited enhanced sensitivity towards VOCs compared with PL and CD detection (Figure 4d), which exhibited the potential in environmental monitoring.

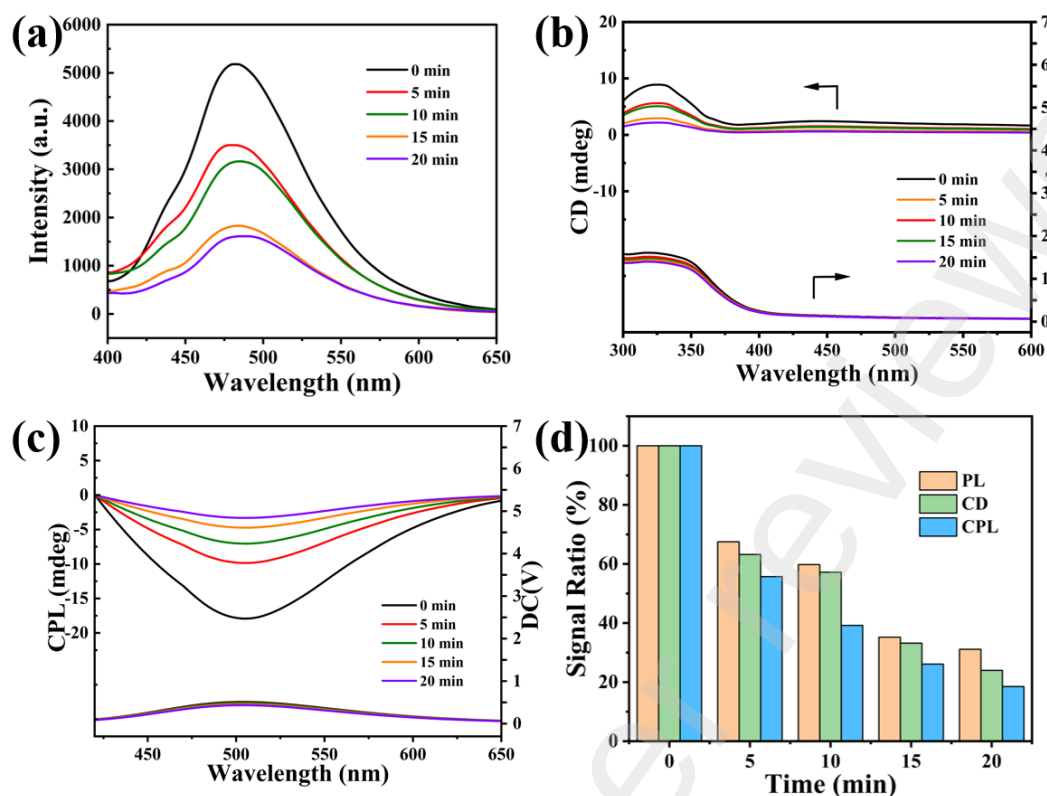


Figure 4. The PL (a), CD (b), CPL spectra (c) of chiral PMMA-based membranes under the fumigation of n-propylamine and the results with time (d).

Conclusion

In this study, a series of multi-armed star-shaped poly(phenylacetylene) derivatives were precisely designed and synthesized, through Rh-complex catalysts. The controllable and living characteristics of polymerization was accomplished on the novel multi-sited initiator NPBA- β -CD. By regulating the feed ratio, the tailor-made PPA-based unimolecular micelles with target molecular weight and size could be obtained with high yields. The β -CD core had the stabilizing effect on the helical structure of conjugated backbones. Compared with linear analogues, star-shaped PPAs displayed the enhanced Cotton effect and larger activation energy of helix inversion. After deprotection, the carboxyl-modified unimolecular micelles were employed as chiral nanoreactors to prepare AIE composites with induced CD and CPL properties, in the form of PMMA-based membranes and CTAB-stabilized suspension. Due to the star-shaped morphology, the enhanced chiral induction of helical unimolecular micelles

was achieved with the comparable CPL activity. Furthermore, this system displayed the triple-mode detection (PL, CD and CPL) towards typical volatile organic compound n-propylamines, which could be further studied for environmental monitoring.

ASSOCIATED CONTENT

Supporting Information

Detailed synthetic procedures of initiators, monomers and polymers. GPC traces of various polymers, along with UV-Vis absorption, CD and CPL spectra. This material is available free of charge *via* the Internet.

AUTHOR INFORMATION

Ge Shi - Henan Joint International Research Laboratory of Living Polymerizations and Functional Nanomaterials, Henan Key Laboratory of Advanced Nylon Materials and Application, School of Materials Science and Engineering, Zhengzhou University, Zhengzhou 450001, China

E-mail: geshizzu@zzu.edu.cn

Xinchang Pang - Henan Joint International Research Laboratory of Living Polymerizations and Functional Nanomaterials, Henan Key Laboratory of Advanced Nylon Materials and Application, School of Materials Science and Engineering, Zhengzhou University, Zhengzhou 450001, China; School of Materials Science and Engineering, Henan University of Science and Technology, Luoyang 471023, China

E-mail: Pangxinchang1980@163.com

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The financial support from the National Science Foundation of China (Grant No. 52273217 to Xinchang Pang; Grant No. 22105179 to Ge Shi; Grant No.52173209 to Yanjie He), Key Scientific Research Projects of Colleges and Universities in Henan Province (Fundamental Research Project 2022) (Grant 22ZX001 to Xinchang Pang), Postdoctoral Foundation of Henan Province (No.304348 to Ge Shi), scientific and

technological research projects in Henan Province (No.242102520004 to Xiaoguang Qiao), and Postgraduate Education Reform and Quality Improvement Project of Henan Province (No. YJS2025GZZ03 to Zhe Cui) are greatly appreciated.

REFERENCES

- (1) Gaoyang Wang, Changlong Hao, Wei Ma, Aihua Qu, Chen Chen, Jun Xu, Chuanlai Xu, Hua Kuang, Liguang Xu. Chiral plasmonic triangular nanorings with sers activity for ultrasensitive detection of amyloid proteins in alzheimer's disease. *Advanced Materials* **2021**, 33 (38), 2102337.
- (2) Weiwei Wang, Jing Zhao, Changlong Hao, Shudong Hu, Chen Chen, Yi Cao, Zhengyu Xu, Jun Guo, Liguang Xu, Maozhong Sun, Chuanlai Xu, Hua Kuang. The development of chiral nanoparticles to target nk cells and cd8⁺ t cells for cancer immunotherapy. *Advanced Materials* **2022**, 34 (16), 2109354.
- (3) Arnold J Boersma, Ben L Feringa, Gerard Roelfes. Enantioselective friedel–crafts reactions in water using a DNA-based catalyst. *Angewandte Chemie International Edition* **2009**, 48 (18), 3346-3348.
- (4) Michael Reggelin, Melanie Schultz, Michael Holbach. Helical chiral polymers without additional stereogenic units: A new class of ligands in asymmetric catalysis. *Angewandte Chemie International Edition* **2002**, 41 (9), 1614-1617.
- (5) Zhenglin Tang, Hiroki Iida, Hai-Yu Hu, Eiji Yashima. Remarkable enhancement of the enantioselectivity of an organocatalyzed asymmetric henry reaction assisted by helical poly(phenylacetylene)s bearing cinchona alkaloid pendants via an amide linkage. *ACS Macro Letters* **2012**, 1 (2), 261-265.
- (6) Chong Zhang, Si Li, Xi-Yan Dong, Shuang-Quan Zang. Circularly polarized luminescence of agglomerate emitters. *Aggregate* **2021**, 2 (5), e48.
- (7) Biao Zhao, Huli Yu, Kai Pan, Zhan'ao Tan, Jianping Deng. Multifarious chiral nanoarchitectures serving as handed-selective fluorescence filters for generating full-color circularly polarized luminescence. *ACS Nano* **2020**, 14 (3), 3208-3218.
- (8) Ming Hu, Feng-Ying Ye, Cong Du, Weizhou Wang, Ting-Ting Zhou, Miao-Li Gao, Minghua Liu, Yan-Song Zheng. Tunable circularly polarized luminescence from single crystal and powder of the simplest tetraphenylethylene helicate. *ACS Nano* **2021**, 15 (10), 16673-16682.
- (9) Hongkun Li, Shan Xue, Huimin Su, Bo Shen, Zhihui Cheng, Jacky W. Y. Lam, Kam Sing Wong, Hongkai Wu, Bing Shi Li, Ben Zhong Tang. Click synthesis, aggregation-induced emission and chirality, circularly polarized luminescence, and helical self-assembly of a leucine-containing silole. *Small* **2016**, 12 (47), 6593-6601.
- (10) Shengzhe Jia, Tiantian Tao, Yujiang Xie, Liuyang Yu, Xiang Kang, Yuan Zhang, Weiwei Tang, Junbo Gong. Chirality supramolecular systems: Helical assemblies, structure designs, and functions. *Small* **2024**, 20 (11), 2307874.
- (11) Yi-Hong Xiao, Zhi-Zhou Ma, Xue-Xian Yang, Dong-Sheng Li, Zhi-Gang Gu, Jian Zhang. Inducing circularly polarized luminescence by confined synthesis of ultrasmall chiral carbon nanodot arrays in pyrene-based mof thin film. *ACS Nano* **2023**, 17 (19), 19136-19143.
- (12) Kevin E. Shopsowitz, Hao Qi, Wadood Y. Hamad, Mark J. MacLachlan. Free-standing mesoporous silica films with tunable chiral nematic structures. *Nature* **2010**, 468 (7322), 422-425.

- (13) Peizhao Liu, Yann Battie, Marion Decossas, Sisareuth Tan, Emilie Pouget, Yutaka Okazaki, Takashi Sagawa, Reiko Oda. Chirality induction to cdse nanocrystals self-organized on silica nanohelices: Tuning chiroptical properties. *ACS Nano* **2021**, 15 (10), 16411-16421.
- (14) Qiao Jiang, Xuehui Xu, Ping-An Yin, Kai Ma, Yonggang Zhen, Pengfei Duan, Qian Peng, Wei-Qiang Chen, Baoquan Ding. Circularly polarized luminescence of achiral cyanine molecules assembled on DNA templates. *Journal of the American Chemical Society* **2019**, 141 (24), 9490-9494.
- (15) Hideyuki Shinmori, Chihiro Mochizuki. Strong chiroptical activity from achiral gold nanorods assembled with proteins. *Chemical Communications* **2017**, 53 (49), 6569-6572.
- (16) V. Sue Myers, Michael G. Weir, Emily V. Carino, David F. Yancey, Surojit Pande, Richard M. Crooks. Dendrimer-encapsulated nanoparticles: New synthetic and characterization methods and catalytic applications. *Chemical Science* **2011**, 2 (9), 1632-1646.
- (17) Minju Kim, Mohammed O. Bazaid, Hyun Jeong Lee, Seung Soon Jang, Yu Jin Jang, Dong Ha Kim. Chiral inorganic nanostructures from achiral platforms: A universal synthesis route via supramolecular self-assembly. *Chemistry of Materials* **2023**, 35 (8), 3073-3082.
- (18) Minju Kim, Jiweon Kim, Hyun Jeong Lee, Hyeohn Kim, Ki Tae Nam, Dong Ha Kim. Block copolymer enabled synthesis and assembly of chiral metal oxide nanoparticle. *ACS Nano* **2023**, 17 (8), 7611-7623.
- (19) Minju Kim, Jiweon Kim, Jieun Bang, Yu Jin Jang, JaeHong Park, Dong Ha Kim. Simultaneously achieving room-temperature circularly polarized luminescence and high stability in chiral perovskite nanocrystals via block copolymer micellar nanoreactors. *Journal of Materials Chemistry A* **2023**, 11 (24), 12876-12884.
- (20) Xinchang Pang, Lei Zhao, Wei Han, Xukai Xin, Zhiquan Lin. A general and robust strategy for the synthesis of nearly monodisperse colloidal nanocrystals. *Nature Nanotechnology* **2013**, 8 (6), 426-431.
- (21) Xinchang Pang, Yanjie He, Jaehan Jung, Zhiquan Lin. 1d nanocrystals with precisely controlled dimensions, compositions, and architectures. *Science* **2016**, 353 (6305), 1268-1272.
- (22) Lingxin Zong, Longwang Kan, Chenrong Yuan, Yanjie He, Wenjie Zhang, Xiaoguang Qiao, Xiaomeng Zhang, Minying Liu, Ge Shi, Xinchang Pang. Chiral confined unimolecular micelles for controlled in situ fabrication of optically active hybrid nanostructures. *The Journal of Physical Chemistry Letters* **2023**, 14 (46), 10361-10368.
- (23) Longwang Kan, Zhenqian Zhang, Junle Zhang, Qianwei Liu, Chenrong Yuan, Yanjie He, Wenjie Zhang, Xiaoguang Qiao, Ge Shi, Xinchang Pang. Precise construction of chiral plasmonic nanoparticles for enantioselective discrimination. *The Journal of Physical Chemistry Letters* **2024**, 15 (30), 7740-7747.
- (24) Ryosuke Sakai, Yoshinobu Mato, Hokuto Ishimaru, Keisuke Ogata, Shota Kurose, Shun Ozawa, Satoshi Umeda, Katsuyuki Tsuda, Toshifumi Satoh, Toyoji Kakuchi. Colorimetric sensing of chirality based on synergistic effect of multiple chiral amide receptors consecutively organized along poly(phenylacetylene) backbone. *Macromolecules* **2024**, 57 (24), 11450-11460.
- (25) Daisuke Hirose, Asahi Isobe, Emilio Quiñoá, Félix Freire, Katsuhiko Maeda. Three-state switchable chiral stationary phase based on helicity control of an optically active poly(phenylacetylene) derivative by using metal cations in the solid state. *Journal of the American Chemical Society* **2019**, 141 (21), 8592-8598.
- (26) Lijia Liu, Yaodong Wang, Fuqingyun Wang, Chunhong Zhang, Yanli Zhou, Zhengjin Zhou,

- Xudong Liu, Ruiqi Zhu, Hongxing Dong, Toshifumi Satoh. Synthesis and asymmetric catalytic performance of one-handed helical poly(phenylacetylene)s bearing proline dipeptide pendants. *Reactive and Functional Polymers* **2020**, 146, 104392.
- (27) Tsuyoshi Taniguchi, Takumi Yoshida, Kensuke Echizen, Kokoro Takayama, Tatsuya Nishimura, Katsuhiro Maeda. Facile and versatile synthesis of end-functionalized poly(phenylacetylene)s: A multicomponent catalytic system for well-controlled living polymerization of phenylacetylenes. *Angewandte Chemie International Edition* **2020**, 59 (22), 8670-8680.
- (28) Kensuke Echizen, Tsuyoshi Taniguchi, Tatsuya Nishimura, Katsuhiro Maeda. Well-controlled living polymerization of phenylacetylenes in water: Synthesis of water-soluble stereoregular telechelic poly(phenylacetylene)s. *Angewandte Chemie International Edition* **2022**, 61 (26), e202202676.
- (29) Chunhong Zhang, Hailun Wang, Qianqian Geng, Taotao Yang, Lijia Liu, Ryosuke Sakai, Toshifumi Satoh, Toyoji Kakuchi, Yoshio Okamoto. Synthesis of helical poly(phenylacetylene)s with amide linkage bearing l-phenylalanine and l-phenylglycine ethyl ester pendants and their applications as chiral stationary phases for hplc. *Macromolecules* **2013**, 46 (21), 8406-8415.
- (30) Shota Mino, Kosuke Matsui, Masahide Goto, Akiyuki Ryoki, Takeyuki Suzuki, Kazushi Fujimoto, Hiromitsu Sogawa, Hiroto Kudo, Fumio Sanda. Star-shaped polymers with helical polyacetylene arms. Comparison of solution- and solid-state properties with linear helical polyacetylenes. *Macromolecules* **2024**, 57 (22), 10824-10834.
- (31) Ayato Inaba, Tatsuya Nishimura, Masato Yamamoto, Sandip Das, Ayhan Yurtsever, Kazuki Miyata, Takeshi Fukuma, Seigo Kawaguchi, Moriya Kikuchi, Tsuyoshi Taniguchi, Katsuhiro Maeda. Synthesis of optically active star polymers consisting of helical poly(phenylacetylene) chains by the living polymerization of phenylacetylenes and their chiroptical properties. *RSC Advances* **2023**, 13 (44), 30978-30984.
- (32) Zhongmin Dong, Zhibin Ye. Synthesis of hyperbranched poly(phenylacetylene)s containing pendant alkyne groups by one-pot pd-catalyzed copolymerization of phenylacetylene with diynes. *Macromolecules* **2012**, 45 (12), 5020-5031.
- (33) Virgil Percec, Jonathan G. Rudick, Mihai Peterca, Martin Wagner, Makoto Obata, Catherine M. Mitchell, Wook-Dong Cho, Venkatachalapathy S. K. Balagurusamy, Paul A. Heiney. Thermoreversible cis-cisoidal to cis-transoidal isomerization of helical dendronized polyphenylacetylenes. *Journal of the American Chemical Society* **2005**, 127 (43), 15257-15264.
- (34) Eiji Yashima, Katsuhiro Maeda, Yoshio Okamoto. Memory of macromolecular helicity assisted by interaction with achiral small molecules. *Nature* **1999**, 399 (6735), 449-451.
- (35) Fumio Sanda, Hitoshi Araki, Toshio Masuda. Synthesis and properties of serine- and threonine-based helical polyacetylenes. *Macromolecules* **2004**, 37 (23), 8510-8516.
- (36) Kevin K. L. Cheuk, Jacky W. Y. Lam, Junwu Chen, Lo Ming Lai, Ben Zhong Tang. Amino acid-containing polyacetylenes: Synthesis, hydrogen bonding, chirality transcription, and chain helicity of amphiphilic poly(phenylacetylene)s carrying l-leucine pendants. *Macromolecules* **2003**, 36 (16), 5947-5959.
- (37) Takahiro Sato, Ken Terao, Akio Teramoto, Michiya Fujiki. On the composition-driven helical screw-sense inversion of chiral-achiral random copolymers. *Macromolecules* **2002**, 35 (14), 5355-5357.
- (38) Katsuhiro Maeda, Mai Nozaki, Kengo Hashimoto, Kouhei Shimomura, Daisuke Hirose, Tatsuya Nishimura, Go Watanabe, Eiji Yashima. Helix-sense-selective synthesis of right- and left-handed

- helical luminescent poly(diphenylacetylene)s with memory of the macromolecular helicity and their helical structures. *Journal of the American Chemical Society* **2020**, *142* (16), 7668-7682.
- (39) Hong-Ru Fu, Dan-Dan Ren, Kun Zhang, Hong Chen, Xiaoyan Lu, Qing-Rong Ding, Lu-Fang Ma. Induced absolute configuration of achiral tetradentate ligands in metal–organic frameworks for circularly polarized luminescence. *Chinese Journal of Chemistry* **2024**, *42* (11), 1260-1266.
- (40) Yan Shi, Guangqiang Yin, Zhiping Yan, Peng Sang, Minghui Wang, Robert Brzozowski, Prahathees Eswara, Lukasz Wojtas, Youxuan Zheng, Xiaopeng Li, Jianfeng Cai. Helical sulfono- γ -aa-peptides with aggregation-induced emission and circularly polarized luminescence. *Journal of the American Chemical Society* **2019**, *141* (32), 12697-12706.
- (41) Biao Zhao, Kai Pan, Jianping Deng. Intense circularly polarized luminescence contributed by helical chirality of monosubstituted polyacetylenes. *Macromolecules* **2018**, *51* (18), 7104-7111.
- (42) Ming Pan, Ran Zhao, Biao Zhao, Jianping Deng. Two chirality transfer channels assist handedness inversion and amplification of circularly polarized luminescence in chiral helical polyacetylene thin films. *Macromolecules* **2021**, *54* (11), 5043-5052.
- (43) Guanghui Ouyang, Jessica Rühe, Yang Zhang, Mei-Jin Lin, Minghua Liu, Frank Würthner. Intramolecular energy and solvent-dependent chirality transfer within a binol-erylene heterocyclophane. *Angewandte Chemie International Edition* **2022**, *61* (31), e202206706.
- (44) Lei Xu, Bao-Rui Gao, Xun-Hui Xu, Li Zhou, Na Liu, Zong-Quan Wu. Controlled synthesis of cyclic-helical polymers with circularly polarized luminescence. *Angewandte Chemie International Edition* **2022**, *61* (28), e202204966.
- (45) Xinmin Zhang, Zhigang Xue, Hong Li, Li Yan, Yuan Yang, Yi Wang, Jingchun Duan, Lei Li, Fahe Chai, Miaomiao Cheng, Weiqi Zhang. Ambient volatile organic compounds pollution in china. *Journal of Environmental Sciences* **2017**, *55*, 69-75.
- (46) Meggie Hakim, Yoav Y. Broza, Orna Barash, Nir Peled, Michael Phillips, Anton Amann, Hossam Haick. Volatile organic compounds of lung cancer and possible biochemical pathways. *Chemical Reviews* **2012**, *112* (11), 5949-5966.